

PBHLTH 245 Final Project Proposal

Mira Terdiman & Angela Dimaano

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Introduction

Our project aims to understand what specific risk factors are associated with six health outcomes: arthritis, cancer, hypertension, diabetes, obesity, and asthma, among a synthetic population of individuals aged 10-89. Specifically, we are curious to explore how both clinical factors (i.e. BMI, Blood Pressure) as well as lifestyle choices (alcohol consumption, diet) impact health. Additionally, we want to better understand whether or not various clinical & lifestyle predictors contribute to underlying health groupings, and whether or not these groupings are related to the six health outcomes listed above.

Research Question

Can we group individuals into underlying/latent health categories based on continuous clinical & lifestyle risk factors? Do these latent health categories have a significant association with the health outcomes of arthritis, cancer, hypertension, diabetes, obesity, and asthma?

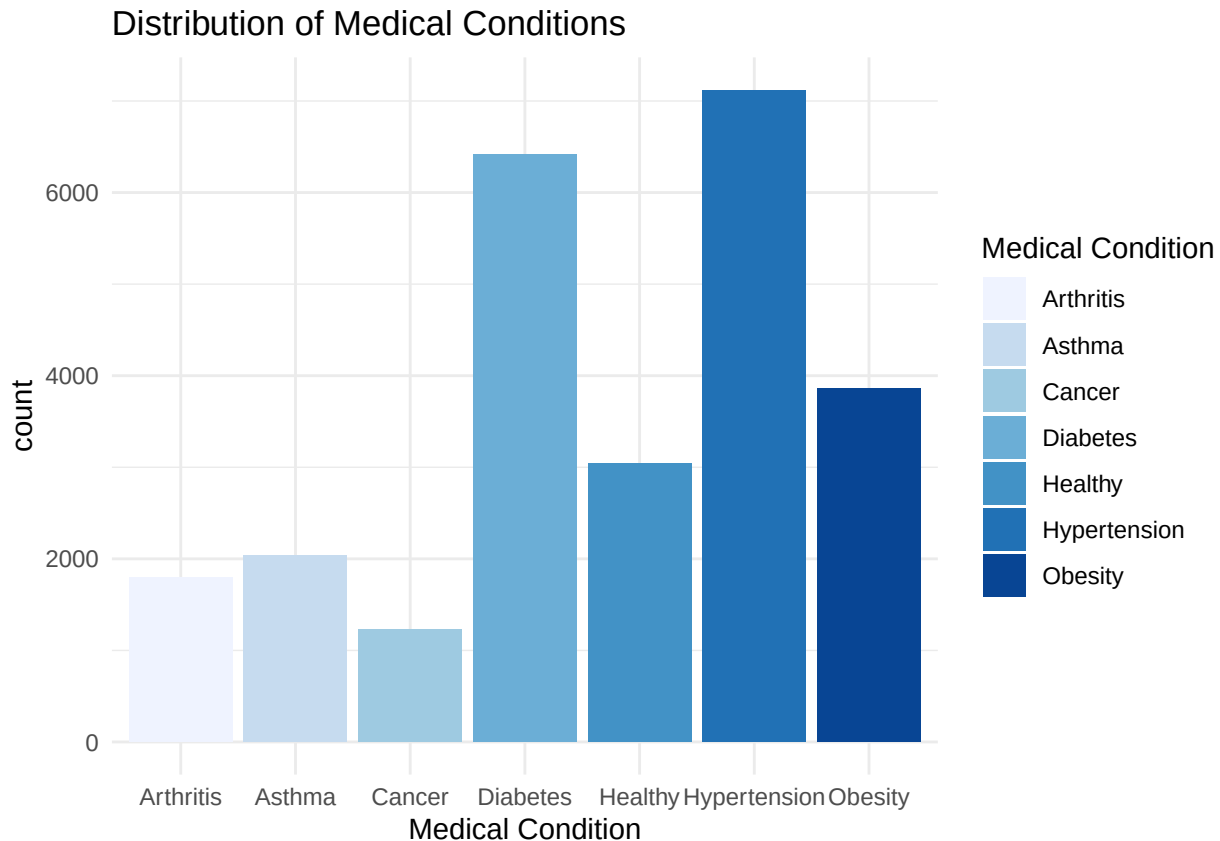
Data

For this project, we are using a dataset pulled from Kaggle titled “Healthcare Risk Factors Dataset,” published by Abdalluh Ahmed. This dataset contains 30,000 entries with 20 variables relating to individuals’ health conditions, including blood pressure, cholesterol, smoking status, BMI, and stress levels. The synthetic dataset was intentionally generated with missing values and noise to allow users to clean and pre-process the data before modeling. We believe this dataset is appropriate for this project given its complexity, as well as its connection to public health. A list of available variables is shown below:

## [1]	"Age"	"Gender"	"Medical Condition"
## [4]	"Glucose"	"Blood Pressure"	"BMI"
## [7]	"Oxygen Saturation"	"LengthOfStay"	"Cholesterol"
## [10]	"Triglycerides"	"HbA1c"	"Smoking"
## [13]	"Alcohol"	"Physical Activity"	"Diet Score"
## [16]	"Family History"	"Stress Level"	"Sleep Hours"
## [19]	"random_notes"	"noise_col"	

Since this dataset provides numerous predictors, we want to see whether or not we can group individuals into underlying classes based on similarities across risk factors. Our outcome of interest is to better understand what groupings of risk factors are associated with the outcomes:

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## Loading required package: viridisLite
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Analysis Plan

To explore whether individuals can be meaningfully grouped into underlying health categories based on their clinical and lifestyle characteristics, we will use a Latent Class Analysis (LCA). Our dataset contains both continuous variables (e.g., BMI, blood pressure, cholesterol) and categorical or ordinal variables (e.g., smoking status, alcohol intake, diet quality), which makes LCA more appropriate than a clustering method like k-Means.

Data Pre-processing

In order to conduct LCA, we will need to clean and transform our data by:

- Either remove or impute missing values (depending on whether or not missingness is random)
- Identify any outliers in dataset, and remove outliers that are not biologically plausible
- Standardize all continuous predictors for interpretability

Latent Class Analysis

LCA will be used to identify unobserved subgroups of individuals who have similar risk profiles. We can fit LCA models with various numbers of classes, and select the model of best fit based on a chosen criterion (i.e. AIC). We can then assign each individual to their most probable class.

Comparing Class Membership to Health Outcomes

After exploring the specific features of each latent risk profile, we will compare classes based on the likelihood of having one of the six health outcomes mentioned above. We will conduct this analysis using regression (health

outcome as the response variable, and latent classes as the predictors), with either logistic or multinomial regression.